MILITARY SPECIFICATION

CABLE ASSEMBLY, SPECIAL PURPOSE, ELECTRICAL, BRANCHED W-K

1. SCOPE

1.1 This specification covers requirements, quality assurance provisions, and the preparation for delivery criteria for one type of cable assembly entitled as Cable Assembly, Special Purpose, Electrical, Branched: W-1-K, W-2-K, W-3-K and W-4-K which are used in the Honest John warhead section.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-810	- Environmental Test Methods.
MIL-STD-1168	- Lot Numbering of Ammunition.
MIL-STD-1460	- Soldering of Electrical Connections and Printed Wiring Assemblies, Procedure For.

DRAWINGS

ORDNANCE CORPS

8800894	- Cable Assembly, Special Purpose, Electrical, Branched: W-4-K.
8808362	- Container, Packing, Ammunition, For Domestic (CONUS) Shipments of Special Weapon Items. Packing and Marking.
8813376	- Cable Assembly, Special Purpose, Electrical, Branched: W-1-K.
8813383	- Cable Assembly, Special Purpose, Electrical, Branched: W-2-K.
8813384	- Cable Assembly, Special Purpose, Electrical, Branched: W-3-K.

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8822033 - Bag, Packing, Ammunition, For Cable Assembly, Special Purpose, Electrical, Branched: W-1-K.
8822034 - Bag, Packing, Ammunition, For Cable Assembly, Special Purpose, Electrical, Branched: W-2-K.
8822035 - Bag, Packing, Ammunition, For Cable Assembly, Special Purpose, Electrical, Branched: W-3-K.
8822041 - Bag, Packing, Ammunition, For Cable Assembly, Special Purpose, Electrical, Branched: W-4-K.
8825153 - Adaption Kit, Atomic Warhead, To Rocket: M86.
EL-8842568 - Equipment List.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The cable assemblies shall comply with all requirements specified on the drawing (dwg) 8800894, 8813383, 8813384 and 8813376, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.4 Serialization. A six letter serial code shall be assigned to the item. Serial codes required on these items shall be obtained by means of a letter addressed to: Commander, Picatinny Arsenal, Dover, New Jersey 07801, ATTENTION: SARPA-QA-T-M. The letter of request shall include:

a. Exact nomenclature of the item as specified on the applicable drawing.
b. Name of the contractor.
c. Contract number or procurement request order number.
d. Number of serial codes required (unused code numbers shall be returned).

3.5 Mating assembly. The cable plugs shall be capable of being fully mated and unmated to a receptacle by hand, without use of tools when tested as specified in 4.4.1.

3.6 Continuity. The resistance of each circuit shall not exceed the value stated for the respective circuit in Tables I through IV when determined as specified in 4.4.2.

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Table I

Cable assembly, W-1-K continuity test

Circuit		
From	To	Maximum Resistance Ohms
P1-A	P2-A	0.04
P1-B	E1-1	0.02
P1-C	E1-1	0.02
P1-D	E1-3	0.02
P1-E	E1-5	0.02
P1-F	E1-6	0.02
P1-G	E1-6	0.02
P2-B	E1-2	0.02
P2-C	E1-2	0.02
P2-D	E1-3	0.02
P2-E	E1-5	0.02
P2-F	E1-4	0.02
P2-G	E1-4	0.02

Refer to Ordnance Drawing No. 8813376 for connector identifications.

Table II

Cable assembly, W-2-K continuity test

Circuit		
From	To	Maximum Resistance Ohms
P1-A	P4-G	0.05
P1-B	P2-k	0.05
P1-C	P2-J	0.05
P1-D	P2-X	0.05
P1-E	P2-H	0.05
P1-F	P4-T	0.05
P1-G	P2-p	0.05
P1-H	P2-d	0.05
P1-J	P2-c	0.05
P1-K	P2-T	0.05
P1-L	P2-A	0.05
P1-M	P2-B	0.05
P1-N	P2-C	0.05
P1-P	P4-Z	0.05
P1-T	P4-G	0.05
P1-U	P2-M	0.05
P1-V	P4-T	0.05
P1-W	P2-i	0.05

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P1-X	P4-X	0.05
P1-Y	P4-W	0.05
P5-A	P3-A	0.05
P5-B	P3-B	0.05
P5-C	P3-E	0.05
P5-D	P3-C	0.05
P5-F	P3-D	0.05
P5-G	P3-J	0.05
P3-F	P6-A	0.05
P3-G	P6-B	0.05
P3-I	E4-1	0.04
P3-K	E2-1	0.04
P3-L	P3-2	0.04
P3-M	E3-1	0.04
P3-N	E4-2	0.04
P4-A	P2-E	0.05
P4-B	P2-G	0.05
P4-C	P2-K	0.05
P4-C	P2-L	0.05
P4-D	P2-P	0.05
P4-E	P2-S	0.05
P4-F	P2-V	0.05
P4-F	P2-W	0.05
P4-Y	P2-D	0.05
P3-H	P6-C	0.05

Refer to Ordnance Drawing No. 8813383 Sheets 1 and 2 for connector identifications.

Table III

Cable assembly, W-3-K continuity test

Circuit		
From	To	Maximum Resistance Ohms
P1-A	P2-A	0.05
P1-B	P2-C	0.05
P1-C	P2-E	0.05
P1-D	P2-F	0.05
P1-E	P2-B	0.05
P1-F	E2-2	0.04
P1-G	E2-3	0.04
P1-H	E3-3	0.04
P1-J	E4-3	0.04
P1-K	P3-C	0.05
P1-L	P2-G	0.05
P1-M	P3-A	0.05

Refer to Ordnance Drawing No. 8813384 for connector identification.

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Table IV

Cable assembly, W-4-K continuity test

Circuit		
From	To	Maximum Resistance Ohms
T1	T2	0.02

3.7 Dielectric strength (W-2-K cable only). The cable shall withstand the application of 1500 ± 200 volts root mean square (vrms), 60 Hertz (Hz) applied from terminals W2KP2-k,p, to W2KP1 - a, b, attachment bushings and connector shells, for 1 minute maximum (max). Testing shall be as specified in 4.4.3.

3.8 Insulation resistance. The insulation resistance of each terminal to every other terminal listed for the respective assemblies in Tables V through VII shall be greater than 100 megohms when tested at 500 ± 50 volts direct current (vdc). Testing shall be as specified in 4.4.4.

Table V

Cable assembly, W-1-K

Insulation resistance test

Contact
P1-A
P1-B
P1-D
P1-E
P1-F
P2-C
P2-G
P1 - Unused contacts, H, J, K - Note 1

Contact
P2 - Unused contacts, H, J, K - Note 1
Ground - This will be the test fixture ground - Note 2.

Note 1 - Unused contacts can be connected together and tested as one contact.
Note 2 - Cable connector shells shall be at ground potential. Refer to Ordnance Drawing No. 8813376 for connector.

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Table VI

Cable assembly, W-2-K

Insulation resistance test

Contact
P1-B
P1-C
P1-D
P1-E
P1-G
P1-H
P1-J
P1-K
P1-L
P1-M
P1-N
P1 - Unused contacts R,S,Z,c - Note 1
P1-U
P1-W
P3-A
P3-B
P3-C
P3-D
P3-E
P3-F
P3-G
P3-H
P3-J
P3-K
P3-I
P3-L
P3-M
P3-N
P4-A
P4-B
P4-C
P4-D
P4-E
P4-F
P4-G
P4 - Unused contacts H,J,K,L,M,N,Q,R,S,U,V - Note 1
P4-T
P4-W
P4-X
P4-Y
P4-Z
P2 - Unused contacts Y,Z,a,b,e,f,g,h,m,n,q,F,J,N,R,U, r,s,t, - Note 1

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P5 - Unused contacts E,H,J,K
Ground - This will be the test fixture ground - Note 2

Note: 1 - Unused contacts can be connected together and tested as one contact.
Note: 2 - Cable connector shells P1-a, P1-b, and attachment bushings shall be ground potential.

Refer to Ordnance Dwg 8813383 Sheets 1 and 2 for connector identifications.

Table VII

Cable assembly, W-3-K

Insulation resistance test

Contact

P1-A
P1-B
P1-C
P1-D
P1-E
P1-F
P1-G
P1-H
P1 - Unused contacts I, N - Note 1
P2 - Unused contacts D, H, J, K - Note 1
P3 - Unused contact B - Note 1
P1-J
P1-K
P1-L
P1-M
Ground - This will be the test fixture ground - Note 2

Note 1 - Unused contacts can be connected together and tested as one contact.
Note 2 - Cable connector shells and attachment bushing shall be at ground potential.

Refer to Ordnance Dwg No. 8813384 for connector identification.

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3.9 Environmental testing. The cable assemblies shall comply with MIL-STD-810, visual inspection and failure criteria, section 3. The cable assemblies shall also comply with 3.6, 3.7 (W-2-K only) and 3.8 before and after each environmental test. Testing shall be as specified in 4.4.5. Tests shall be conducted in the order specified.

3.9.1 Temperature shock. The cable assemblies with their respective plugs mated to matching receptacles shall withstand temperature shock without damage, and shall meet the requirements of 3.6, 3.7 (W-2-K only) and 3.8 after testing as specified in 4.4.5.1.

3.9.2 Humidity. The cable assemblies with their respectiv plugs mated to matching receptacles shall withstand exposure to a relative humidity of 95 plus or minus (±) 5 percent at temperatures cycling from 80 ± 20 degrees Fahrenheit (°F), to 160 ± 5°F to 80 ± 20°F without evidence of cracking, peeling or any other damage when tested as specified in 4.4.5.2. Upon completion of this test the cable assemblies shall comply with 3.6, 3.7 (W-2-K only) and 3.8.

3.9.3 Low temperature storage. The cable assemblies with their respective plugs mated to matching receptacles shall withstand exposure to low temperature storage of -80 ± 5°F without cracking, chipping, shrinkage, embrittlement, distortion, or other damage and shall meet the requirements of 3.6, 3.7 (W-2-K only) and 3.8 while at -65 ± 5°F when tested as specified in 4.4.5.3.

3.9.4 Vibration. The cable assemblies shall withstand exposure to vibration in two mutually perpendicular axes at the frequencies, amplitudes and temperatures specified in Table VIII for the times specified in Table VIII when tested as in 4.4.5.4. Upon completion of this test the cable assemblies shall meet the requirements of 3.6,3.7 (W-2-K only) and 3.6.

Table VIII

Frequency Range Hertz (Hz)	Double Displacement or Peak g Level ± 10 Percent	Cycle Time to Cover Frequency Range (Minutes)	Minutes Test Time Each Axis ± 10 Percent 80 ± 20°F	160 ± 5°F	-65 ±
10-20-10	± 2 g	3 ± 0.5	30	10	10
20-50-20	0.036 inch	3 ± 0.5	15	10	10
50-500-50	± 5 g	3 ± 0.5	15	10	10

3.9.5 Shock.- The cable assemblies shall withstand exposure to shocks of 75 ± 8 g's at the temperatures and for the times specified in Table IX without evidence of cracking or other damage and shall maintain continuity through the circuits specified in Table IX during shocks when tested as specified in 4.4.5.5.

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Table IX

Peak Shock Amplitude	Time Duration Half Amplitude	Temperature
75 ± 8 g's	9 ± 3 milliseconds	-65 ± 5°F
75 ± 8 g's	9 ± 3 milliseconds	80 ± 20°F
75 ± 8 g's	9 ± 3 milliseconds	165 ± 5°F

Table X

From Pin	To Pin	From Pin	To Pin
W1KP1-B	E1-1	W2KP3-F	W2KP6-A
W1KP1-F	E1-6	W3KP3-A	W3KP1-M
W1KP2-F	E1-4	W3KP1-F	E2-2
W2KP3-K	E2-1	W3KP1-H	E3-3
W2KP3-M	E3-1	W3KP1-J	E4-3
W2KP3-I	E4-1	W4KT1	W4KT2
W2KP2-B	E1-2		

3.10 Soldering. All soldering shall comply with the requirements specified in MIL-STD-1460.

3.11 Workmanship.

3.11.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.11.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.11.3 Burrs. Parts shall be free from burrs.

3.11.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.11.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

3.11.6 Electrical. Cable assemblies and wires shall be dressed in such a manner as to prevent pinching or chafing of the wire insulation. Ends of wires shall be free from burrs, sharp edges, or points. All parts shall be free from flux, chips, dirt, grease, rust or foreign material.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own

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or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.

4.2.1 Quantity to be submitted. When the contract requires first article inspection, the contractor shall submit 5 complete assemblies (see dwgs 8800894, 8813383, 8813384 and 8813376) for first article testing to the testing facility designated in the invitation for bids or request for proposal.

4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in Table XI. Tests shall be conducted in order specified in Table XI.

Table XI

First article inspection and tests

Inspection or Test	W-1-K	W-2-K	W-3-K	W-4-K	Dest	Non-Dest
1. Classification of characteristics (see 4.3.2.1)	X	X	X	X		X
2. Mating assembly (see 3.5)	X	X	X	X		X
3. Continuity (see 3.6)	X	X	X	X		X
4. Dielectric strength (see 3.7)		X				X
5. Insulation resistance (see 3.8)	X	X	X	X		X
*6. Vibration (see 3.9.4)	X	X	X	X	X	
*7. Shock (see 3.9.5)	X	X	X	X	X	
*8. Temperature shock (see 3.9.1)	X	X	X	X	X	
*9. Low temperature storage (see 3.9.3)	X	X	X	X		X
*10. Humidity (see 3.9.2)	X	X	X	X	X	

*Sequence of these tests is optional.

4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Quality conformance inspection.

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4.3.1 Lot formation.

4.3.1.1 Production lot. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification and process changes not affecting safety, performance, storage, or interchangeability, as determined by the Government, shall not necessitate changing the lot interfix number. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

4.3.1.2 Inspection lot. A lot or batch and its formation, size, and presentation is described in MIL-STD-105, Section 5. Accordingly, a lot shall mean an inspection lot and a batch shall mean an inspection batch for the purpose of this specification. The manner in which each inspection lot or batch is to be presented and identified by the contractor, shall be designated or approved by the Government representative.

4.3.2 Product inspection examination. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table (see Table XII and Table XIII), and inspected for all characteristics in the applicable classification. Any characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

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Table XII
Major Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-72	all	140-141	124	256-259	177	541-549	230
73	72	142	125	260-262	178	550-559	231
74	73	143-144	126	263-265	179	560-569	232
75	74	145	127	266-268	180	570-579	233
76	75	146-147	128	269-271	181	580-590	234
77	76	148-149	129	272-275	182	591-601	235
78	77	150	130	276-278	183	602-613	236
79	78	151-152	131	279-282	184	614-625	237
80	79	153-154	132	283-285	185	626-637	238
81-82	80	155-156	133	286-289	186	638-649	239
83	81	157	134	290-293	187	650-662	240
84	82	158-159	135	294-296	188	663-676	241
85	83	160-161	136	297-300	189	677-690	242
86	84	162-163	137	301-304	190	691-704	243
87	85	164-165	138	305-308	191	705-719	244
88-89	86	166-167	139	309-312	192	720-735	245
90	87	168	140	313-316	193	736-751	246
91	88	169-170	141	317-320	194	752-768	247
92	89	171-172	142	321-324	195	769-785	248
93	90	173-174	143	325-329	196	786-804	249
94-95	91	175-176	144	330-333	197	805-823	250
96	92	177-178	145	334-338	198	824-842	251
97	93	179-180	146	339-342	199	843-863	252
98	94	181-182	147	343-347	200	864-884	253
99-100	95	183-184	148	348-352	201	885-907	254
101	96	185-187	149	353-356	202	908-930	255
102	97	188-189	150	357-361	203	931-955	256
103	98	190-191	151	362-366	204	956-981	257
104-105	99	192-193	152	367-372	205	982-1008	258
106	100	194-195	153	373-377	206	1009-1036	259
107	101	196-197	154	378-382	207	1037-1066	260
108-109	102	198-200	155	383-388	208	1067-1098	261
110	103	201-202	156	389-393	209	1099-1131	262
111	104	203-204	157	394-399	210	1132-1166	263
112-113	105	205-207	158	400-405	211	1167-1204	264
114	106	208-209	159	406-411	212	1205-1243	265
115	107	210-212	160	412-417	213	1244-1285	266
116-117	108	213-214	161	418-424	214	1286-1330	267
118	109	215-217	162	425-430	215	1331-1378	268
119	110	218-219	163	431-437	216	1379-1429	269
120-121	111	220-222	164	438-443	217	1430-1483	270
122	112	223-224	165	444-450	218	1484-1542	271
123-124	113	225-227	166	451-457	219	1543-1605	272
125	114	228-230	167	458-465	220	1606-1673	273
126-127	115	231-232	168	466-472	221	1674-1747	274
128	116	233-235	169	473-480	222	1748-1827	275
129-130	117	236-238	170	481-488	223	1828-1914	276
131	118	239-241	171	489-496	224	1915-2010	277
132-133	119	242-243	172	497-504	225	2011-2115	278
134	120	244-246	173	505-513	226	2116-2232	279
135-136	121	247-249	174	514-521	227	2233-2361	280
137	122	250-252	175	522-530	228	over 2361	298
138-139	123	253-255	176	531-540	229		

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Table XIII
Minor Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

4.3.2.1 Classification of characteristics (see 3.2). Critical, major and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Cable assemblies (see dwgs 8800894, 8813376, 8813383, and 8813384 as applicable) before molding.

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Any components missing, damaged or loose	Visual
	102. Improper connector	Visual
	103. Presence of chips or foreign matter unless removable	Visual
	104. Evidence of poor workmanship (see 3.11)	Visual
	105. Improper soldering (see 3.10)	Visual
	106. Objective evidence relating to material missing or improper (see 3.1)	Visual
Minor:	None defined.	

4.3.2.1.2 Cable assemblies (see dwgs 8800894, 8813376, 8813383, and 8813384, as applicable) after molding.

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Length improper	Gage
	102. Foreign matter on inserts unless removable	Visual

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103.	Any pin bent	Visual
104.	Foreign matter on pin contacts or in socket contacts	Visual
105.	Any pin missing (as evidenced by empty hole in insert)	Visual
106.	Rubber on face of bushings	Visual
107.	Voids or air pockets in molded rubber greater than 1/8 inch in diameter	Visual
108.	Leadwire not completely covered by molded rubber	Visual
*109.	Depth of sockets (female sockets of the same diameter must all appear to be within 1/16 inch in depth relative to the adjacent sockets)	Visual
**110.	Height of pins (pins of the same diameter must all appear to be within 1/16 inch in height relative to the adjacent pins)	Visual
111.	Evidence of poor workmanship (see 3.11)	Visual
112.	Any component missing or insecurely assembled	Visual/Manual
113.	Deformation of connector shell sufficient to impair mating	Visual
114.	Any component cracked	Visual
115.	Portion of connector broken away	Visual
116.	Objective evidence relating to material missing or improper (see 3.1)	Visual

Minor:

201.	Marking mislocated, incorrect or unidentifiable	Visual
202.	Lack of mechanical bonding between the molded rubber and inserts, connectors, bushings, and rubber tubing	Visual
203.	Serialization improper or missing (see 3.4)	Visual

*In the event socket contacts do not appear to be within 1/16 inch in depth relative to adjacent sockets, a gage (see 4.3.4.1) shall be used to determine that variation from the normal depth. Reject any insert that varies by more than 0.063 inch maximum from the normal socket depth of the individual connector. The normal depth of the individual connector sockets shall be determined by calculating the average socket that appears to be defective. The forward edge of the connector shall be utilized as a reference surface for all measurements of socket depths.

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**In the event pins do not appear to be within 1/16 inch in height relative to the adjacent pins, a gage (see 4.3.4.1) shall be used to determine that variation from the normal height. Reject any pin that varies by more than 0.063 inch maximum from the normal pin height of the individual connector. The normal height of the individual connector pens shall be determined by calculating the average height from measurements taken of pins adjacent to the pen that appears to be defective. The forward edge of the connector shell shall be utilized as the reference surface for all measurements of pen heights.

4.3.2.1.3 Packing (bag) cable assemblies (see dwgs 8822033 8822034 8822035 and 8822041).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Wrong size bag	Visual
102.	Bag not sealed	Visual
103.	Air in bag not minimized before sealing	Visual
104.	Marking incorrect, missing or mislocated	Visual
105.	Evidence of poor workmanship (see 3.11)	Visual
106	Objective evidence relating to material missing or improper (see 3.1)	Visual
Minor:	None defined.	

4.3.2.1.4 Packing (container) cable assemblies (see dwg. 8808362).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
101.	Marking incorrect, missing, or mislocated	Visual
102.	Packing material visibly wet	Visual
103.	Incorrect or insufficient cushioning material	Visual
104.	Evidence of poor workmanship (see 3.11)	Visual

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105. Objective evidence relating to material missing or improper (see 3.1) .....Visual

Minor: None defined.

4.3.3 Testing. All tests shall be performed. Tests shall be performed in the order specified below:

	Test	Test classification
a.	Mating assembly	Major
b.	Continuity	Major
c.	Dielectric strength	Major
d.	Insulation resistance	Major
e.	Vibration	Major

Of these tests, e is considered destructive and samples so tested shall not be returned to the lot. Destructive samples are over and above the deliverable quantity of items required by the contract.

4.3.3.1 Mating assembly, continuity, dielectric strength (W-2-K only) and insulation resistance. Each assembly in the lot shall be tested in accordance with 4.4.1 through 4.4.4. If any item fails to comply with 3.5 through 3.8, the item shall be rejected and removed from the lot.

4.3.3.2 Vibration. A sample shall be selected at random in accordance with Table IV. The sample shall be tested in accordance with 4.4.5.4. If any item fails to comply with 3.9.4, the lot shall be rejected.

Table IV

Sample sizes for Destructive testing

Number Required to be Delivered	Sample Size for No Failures
2-8	1
9-180	2
181-500	3
501-800	5
801-1300	7

Lot Size Required = Number Required to be Delivered + Sample Size.

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4.3.4 Acceptance inspection equipment.

4.3.4.1 Major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL) (see EL-8842568). When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment and obtain approval prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired (see 6.3).
b. The code "CD" is listed in the "Part or Identifying Number" column on the EL (see 6.1.2).

4.3.4.2 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on ELs. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

4.4.1 Mating assembly. The respective cable plugs shall be mated to matching receptacles. Care must be exercised to avoid bending connector pens or sockets.

4.4.2 Continuity. The resistance shall be measured by means of a suitable electrical resistance measuring instrument which has been connected to the cable assembly contacts specified.

4.4.3 Dielectric strength (W-2-K cable only). The test voltage shall be applied by an instrument capable of supplying the required voltage. The test instrument shall be current limited to five milliamperes maximum or adequately interlocked to protect human life.

4.4.4 Insulation resistance. An insulation resistance tester, capable of supplying the required voltage and with which a resistance in megohms can be obtained, shall be used to measure the resistance. The current shall be limited to a maximum of 5 milliamperes to protect human life.

4.4.5 Environmental testing (see 3.9)

4.4.5.1 Temperature shock. The assembly shall be placed within a test chamber where the temperature is -65 ± 5°F. The assembly shall be subjected to this temperature for a period of 4 hours minimum at the conclusion of which, and within 5 minutes, the assembly shall be transferred to a chamber having an internal temperature of 160 ± 5°F.

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The item shall be subjected to this temperature for a period of 4 hours minimum. This completes one cycle. The number of complete cycles shall be three with a maximum of 5 minutes transfer time between test chambers. At the conclusion of the third cycle the item shall be removed from the test chamber and within a period of 30 minutes shall be tested as specified in 4.4.2, 4.4.3 and 4.4.4.

4.4.5.2 Humidity. The assembly shall be placed in a humidity test chamber. The chamber temperature shall be 80 ± 20°F. During the first 2 hour period, the temperature shall be gradually raised to 160 ± 5°F. The temperature of 160 ± 5°F shall be maintained during the next 6-hour period. During the following 16 hour period, the temperature in the chamber shall be gradually reduced to 80 ± 20°F which completes one cycle. The relative humidity throughout the cycle shall be 95 + 5 percent. The test shall consist of 15 such cycles (360 hours). Upon completion of the fifteenth cycle, the assembly shall be removed immediately from the test chamber and excessive moisture removed by blowing dry, filtered, compressed air over the unit for a period of 5 minutes. The assembly shall then be tested as specified in 4.4.2, 4.4.3 and 4.4.4.

4.4.5.3 Low temperature storage. The cable assembly shall be coiled so that the diameter of the coil is not more then 12 inches. The coiled cable shall then be placed in a chamber and the chamber temperature shall be lowered to -80 ± 5°F. The cable assembly shall remain at this temperature for a minimum of 4 hours. The chamber temperature shall then be raised to -65 ± 5°F for an additional 1 hour period or until stabilization (see 6.4) is reached, whichever is longer. At the end of the 1 hour period and while still exposed to this temperature, the cable assembly shall be immediately uncoiled to its full length and tested as specified in 4.4.2, 4.4.3 and 4.4.4.

4.4.5.4 Vibration. The cable assembly shall be attached to a fixture that provides attachment points and cable terminations in the same positions as shown on dwg 8825153. The cable assembly shall be vibrated at the frequencies, g amplitudes and for the times specified in Table VIII in each of two mutually perpendicular axis. The axis along the lengthwise (long) plane of the cable does not require vibration. Following this test, the cable assembly shall be tested as specified in 4.4.2, 4.4.3 and 4.4.4.

4.4.5.5 Transportation shock. The cable assembly shall be attached to a fixture that provides attachment points and cable terminations in the same positions as shown on dwg 8825153. The fixture shall be mounted to a

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suitable shock producing device and placed in a suitable test chamber. The continuity of the circuit listed in Table X, under the operating conditions given, shall be measured continuously during the test with instrumentation capable of detecting circuit openings of 20 microseconds duration. Following this test, the assembly shall be tested as specified in 4.4.2, 4.4.3 and 4.4.4.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The cable assemblies shall be packaged in accordance with dwg 8822033, 8822034, 8822035 and 8822041 using the specified moisture barrier.

5.1.2 Level C. The cable assemblies shall be packaged in accordance with dwg 8822033, 8822034, 8822035 and 8822041 except that the moisture barrier may be omitted.

5.2 Packing.

5.2.1 Level A and C. The cable assemblies shall be packed in accordance with dwg 8808362.

5.3 Marking. Marking shall be as specified in accordance with dwg 8822033, 8822034, 8822035, 8822041 and 8808362 as applicable.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

a. Title, number, and date of this specification.
b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.1.2 Contract data requirements (see 4.3.4.1). Data specified in 4.3.4.1 will be delivered as identified on a numbered DD Form 1664 when required on a DD Form 1423 incorporated in the contract.

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commander, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SARPA-QA-N-I.

6.4 Temperature stabilization. The temperature is considered stabilized when the greatest peaks (both highest and lowest) continually observed over 5 minutes min. do not exceed the nominal temperature plus tolerance specified.

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6.5 Supersession data. This specification supersedes the requirements of Purchase Description PA-PD-1692 dated 28 July 1961.

Custodian:
Army - PA

Preparing activity:
Army - PA

Project No. 1135-A144

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